Mahavishnu Ponnusamy

AI Full-Stack Engineer | FastAPI, Python · LangChain, LangGraph · Laravel, PHP · React.js
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PROFESSIONAL SUMMARY

AI Full-Stack Engineer with **6+ years** in production web systems and **2+ years** building AI-powered platforms using **FastAPI**, **LangChain**, and **LangGraph**. Shipped an AI legal research platform with multi-agent orchestration, hybrid vector search (**Qdrant**, **Pinecone**), and **RAG** responses, reducing document retrieval latency by **~40%**. Delivered **20+ production applications** across distributed queues (**RabbitMQ**), high-availability Linux environments, and full-stack **React.js/Laravel** architectures.

TECHNICAL SKILLS

- **AI & Retrieval**: LangChain, LangGraph, RAG, OpenAI API, Anthropic, FastMCP, Vector Search (Qdrant), OCR
- **Languages & Core**: Python, PHP, JavaScript, TypeScript, Bash, SQL, OOP
- **Frameworks & UI**: FastAPI, Laravel, React.js, Next.js, Inertia.js, Tailwind CSS
- **Databases & Cache**: PostgreSQL, MySQL, Redis, MongoDB, Qdrant
- **DevOps, Cloud & Testing**: AWS, DigitalOcean, Docker, Nginx, Linux, CI/CD, Pytest, PHPUnit

PROFESSIONAL EXPERIENCE

Hectadata India Technology Consulting Pvt Ltd — ***Remote, Malaysia***

PHP Developer — *Jul 2024 - Present*

- Built an AI legal research platform backend (FastAPI, Python 3.12) using LangGraph multi-agent orchestration, hybrid vector retrieval (Pinecone dense + Qdrant BM25 sparse), and real-time Socket.IO chat, cutting case research latency by **~40%**.
- Architected a clean Onion Architecture in FastAPI with SQLAlchemy 2 async, Pydantic v2, Alembic, MongoDB, and FastMCP server tools, securing **10+ internal user roles**.
- Engineered an end-to-end logistics document extraction pipeline using OCR across DOCX/PDF/images and pluggable LLM routing, processing **500+ documents monthly**.
- Contributed to a real-time vehicle enforcement platform (Laravel 12, RabbitMQ, PostgreSQL NOTIFY) processing **lakhs of records daily (10+ rec/sec)** ensuring **zero data loss and 100% data integrity** across **5+ remote sensor stations**.
- Deployed and maintained distributed services across cloud and on-premises Linux servers using **Nginx load balancing**, reverse proxy, Docker, and Supervisor with **99.9% uptime**.

Byrut Business Solutions LLP — ***Remote, Andaman and Nicobar Islands***

Web Developer — *Nov 2020 - Jun 2024*

- Delivered **20+ production web applications** across tourism, e-commerce, and business automation, driving end-to-end architecture, API integrations, and deployments over 4 years.
- Architected and scaled a multi-service tourism booking platform to **25,000+ monthly users** (hotel, ferry, cab, boat) using Laravel, React.js, and MySQL, hosted on **AWS**.
- Developed dynamic tour package pricing logic and ferry booking engines deployed on **Laravel Cloud**; integrated FPDF generating **1,000+ monthly client quotations** and Razorpay payment gateway.
- Deployed and managed 20+ client applications on **DigitalOcean** droplets via Laravel Forge, utilizing Spaces for object storage and CDN for global asset delivery.
- Built React.js / Next.js administrative portals and multi-vendor e-commerce platforms supporting **30+ active vendors** with scheduled weekly payout workflows.

PROJECTS

- **AI Search Engine —** Enterprise semantic search platform (**FastAPI**, **React**, **Qdrant**, **LangChain**, **RAG**) with **hybrid vector retrieval**, delivering sub-second (**<500ms**) query responses across document collections.
- **WhatsApp Chatbot —** Production chatbot (**FastAPI**, **Meta WhatsApp Cloud API**) for a financial cooperative automating **80%+ of incoming loan FAQs**, human escalation, and broadcast messaging with a **React** admin SPA.

EDUCATION

Paavai Engineering College — ***Namakkal, Tamil Nadu, India***

B.Tech in Information Technology CGPA: 7.75/10 — *Aug 2017 - June 2021*

AWARDS/CERTIFICATION

- **Research Publication:** Published conference paper on *Blockchain-based invoice factoring and secure credit-based training*.
- **2nd Place Winner:** National-level AI technical symposium paper presentation on Artificial Intelligence.
- **Smart India Hackathon:** Selected and participated in **Smart India Hackathon 2020** nationwide finals.